Namespace SixLabors.ImageSharp

Classes

Configuration
: Provides configuration which allows altering default behaviour or extending the library.

GeometryUtilities
: Utility class for common geometric functions.

GraphicOptionsDefaultsExtensions
: Adds extensions that allow the processing of images to the Image<TPixel> type.

GraphicsOptions
: Provides configuration for controlling how graphics operations are rendered, including antialiasing, pixel blending, and alpha composition.

Image
: Encapsulates an image, which consists of the pixel data for a graphics image and its attributes. For the non-generic Image type, the pixel type is only known at runtime. Image is always implemented by a pixel-specific Image<TPixel> instance.

ImageExtensions
: Extension methods for the Image type.

ImageFormatException
: The exception that is thrown when the library tries to load an image, which has format or content that is invalid or unsupported by ImageSharp.

ImageFrame
: Represents a pixel-agnostic image frame containing all pixel data and ImageFrameMetadata. In case of animated formats like gif, it contains the single frame in a animation. In all other cases it is the only frame of the image.

ImageFrameCollection
: Encapsulates a pixel-agnostic collection of ImageFrame instances that make up an Image.

ImageFrameCollectionExtensions
: Extension methods for ImageFrameCollection<TPixel>.

ImageFrameCollection<TPixel>
: Encapsulates a pixel-specific collection of ImageFrame<TPixel> instances that make up an Image<TPixel>.

ImageFrame<TPixel>
: Represents a pixel-specific image frame containing all pixel data and ImageFrameMetadata. In case of animated formats like gif, it contains the single frame in a animation. In all other cases it is the only frame of the image.

ImageInfo
: Contains information about the image including dimensions, pixel type information and additional metadata

ImageMetadataExtensions
: Extension methods for the ImageMetadata and ImageFrameMetadata types.

ImageProcessingException
: The exception that is thrown when an error occurs when applying a process to an image.

Image<TPixel>
: Encapsulates an image, which consists of the pixel data for a graphics image and its attributes. For generic Image<TPixel>-s the pixel type is known at compile time.

IndexedImageFrame<TPixel>
: A pixel-specific image frame where each pixel buffer value represents an index in a color palette.

InvalidImageContentException
: The exception that is thrown when the library tries to load an image which contains invalid content.

Matrix3×2Extensions
: Extension methods for the Matrix3×2 struct.

UnknownImageFormatException
: The exception that is thrown when the library tries to load an image which has an unknown format.

Structs

Color
: Represents a color value that is convertible to any IPixel<TSelf> type.

ColorMatrix
: A structure encapsulating a 5×4 matrix used for transforming the color and alpha components of an image.

DenseMatrix<T>

Represents a dense matrix with arbitrary elements. Components that are adjacent in a column of the matrix are adjacent in the storage array. The components are said to be stored in column major order.

Number
Represents an integral number.

PixelAccessor<TPixel>
Provides efficient access the pixel buffers of an Image<TPixel>.

Point
Represents an ordered pair of integer x- and y-coordinates that defines a point in a two-dimensional plane.

PointF
Represents an ordered pair of single precision floating point x- and y-coordinates that defines a point in a two-dimensional plane.

Rational
Represents a number that can be expressed as a fraction.

Rectangle
Stores a set of four integers that represent the location and size of a rectangle.

RectangleF
Stores a set of four single precision floating points that represent the location and size of a rectangle.

SignedRational
Represents a number that can be expressed as a fraction.

Size
Stores an ordered pair of integers, which specify a height and width.

SizeF
Stores an ordered pair of single precision floating points, which specify a height and width.

Interfaces

IDeepCloneable
An interface for objects that can be cloned. This creates a deep copy of the object.

IDeepCloneable<T>
A generic interface for a deeply cloneable type.

Enums

ByteOrder
: The byte order of the data stream.

ColorHexFormat
: Specifies the channel order when formatting or parsing a color as a hexadecimal string.

ReadOrigin
: Specifies the position in a stream to use for reading.

Delegates

PixelAccessorAction<TPixel>
: A delegate to be executed on a PixelAccessor<TPixel>.

PixelAccessorAction<TPixel1, TPixel2>
: A delegate to be executed on two instances of PixelAccessor<TPixel>.

PixelAccessorAction<TPixel1, TPixel2, TPixel3>
: A delegate to be executed on three instances of PixelAccessor<TPixel>.

Namespace SixLabors.ImageSharp.Drawing

Classes

ArcLineSegment
: Represents a line segment that contains radii and angles that will be rendered as an elliptical arc.

ClipPathExtensions
: Provides extension methods to IPath that allow the clipping of shapes.

ComplexPolygon
: Represents a complex polygon made up of one or more shapes overlayed on each other, where overlaps cause holes.

CubicBezierLineSegment
: Represents a line segment that contains a list of control points that will be rendered as a cubic bezier curve.

EllipsePolygon
: An elliptical shape made up of a single path made up of one or more ILineSegments.

EmptyPath
: A path that is always empty.

LinearGeometry
: Represents retained linearized geometry that can be consumed directly by drawing backends.

LinearLineSegment
: Represents a series of control points that will be joined by straight lines.

NormalizePathExtensions
: Provides extension methods to IPath that allow paths to be normalized.

OutlinePathExtensions
: Extensions to IPath that allow the generation of outlines.

Path
: An aggregate of ILineSegments making a single logical path.

PathBuilder
: Provides a fluent API for incrementally building complex shapes and paths from figures composed of lines, arcs, and bezier curves.

PathCollection
: An aggregate of IPaths to apply common operations to them.

PathExtensions
: Convenience methods that can be applied to shapes and paths.

PiePolygon
: A pie sector polygon defined by a center point, radii, rotation, and arc sweep.

Polygon
: A shape made up of a single closed path made up of one or more ILineSegments.

RectanglePolygon
: A closed rectangular path defined by four straight edges.

Region
: Represents an integer region composed from axis-aligned rectangles.

RegularPolygon
: A regular polygon with a configurable number of equally spaced vertices placed on a circumscribing circle.

RoundedRectanglePolygon
: A closed rectangular path with rounded corners.

SplitPathExtensions
: Extensions to IPath for splitting paths into dash segments without performing stroke expansion.

StarPolygon
: A star-shaped polygon defined by alternating inner and outer radii.

Structs

LinearContour
: Describes a single contour within a LinearGeometry.

LinearGeometryInfo
: Describes geometry-wide metadata for a LinearGeometry instance.

LinearSegment
: Represents one derived linear segment within a LinearGeometry.

PathPoint
: Represents metadata about a point along a path.

SegmentEnumerator
: Enumerates the derived linear segments in a LinearGeometry.

Interfaces

ILineSegment
: Represents a simple path segment.

IPath
: Represents a logic path that can be drawn.

IPathCollection
: Represents a collection of paths that can be enumerated and transformed as a single unit.

ISimplePath
: Represents a simple (non-composite) path defined by a series of points.

Enums

BooleanOperation
: Specifies the type of boolean operation to perform on polygons.

ClipOperation
: Specifies how a clip shape is combined with the current clip region.

IntersectionRule
: Provides options for calculating intersection points.

LineCap
: Specifies the shape to be used at the ends of open lines or paths when stroking.

LineJoin
: Specifies how the connection between two consecutive line segments (a join) is rendered when stroking paths or polygons.

PathTypes
: Describes the different type of paths.

Namespace SixLabors.ImageSharp.Web

Classes

ApplicationBuilderExtensions
: Extension methods for IApplicationBuilder to simplify middleware registration.

CaseHandlingUriBuilder
: Optimized helper methods for generating encoded Uris from URI components. Much of this code has been adapted from the MIT licensed .NET runtime.

ExifOrientationUtilities
: Contains various helper methods for working with EXIF orientation values.

Format
: Contains reusable static instances of known image formats.

FormatUtilities
: Contains various helper methods for working with image formats based on the given configuration.

FormattedImage
: A class encapsulating an image with a particular file encoding.

HMACUtilities
: Provides methods to compute a Hash-based Message Authentication Code (HMAC).

ImageSharpBuilderExtensions
: Extension methods for IImageSharpBuilder that allow configuration of services.

RequestAuthorizationUtilities
: Contains various helper methods for authorizing image requests.

Resampler
: Contains reusable static instances of known resampling algorithms.

ServiceCollectionExtensions
: Extension methods for IServiceCollection to simplify middleware service registration.

Structs

FormatCommand
: Represents a command for setting the type of format with which to encode processed images.

ImageCacheMetadata
: Represents the metadata associated with an image file.

ImageMetadata
: Represents the metadata associated with an image file.

ResamplerCommand
: Represents a command for setting the type of sampler to use for resampling operations.

Interfaces

IImageSharpBuilder
: Defines a contract for adding ImageSharp services.

Enums

CaseHandlingUriBuilder.CaseHandling
: Provides Uri case handling options.

CommandHandling
: Provides enumeration for handling CommandCollection instances when processing a request.

Namespace SixLabors.ImageSharp.Web.Azure.Resolvers

Classes

AzureBlobStorageCacheResolver

Provides means to manage image buffers within the AzureBlobStorageCache.

AzureBlobStorageImageResolver

Provides means to manage image buffers within the Azure Blob file system.

Namespace SixLabors.ImageSharp.Web.AWS

Classes

AmazonS3BucketClient
: Represents a scoped Amazon S3 client instance that is explicitly associated with a single S3 bucket. This wrapper provides a strongly-typed link between the client and the bucket it operates on, and optionally manages the lifetime of the underlying Amazon.S3.AmazonS3Client.

Interfaces

IAWSS3BucketClientOptions
: Provides a common interface for AWS S3 Bucket Client Options.

Namespace SixLabors.Fonts

Classes

Font
: Defines a particular format for text, including font face, size, and style attributes. This class cannot be inherited.

FontCollection
: Represents a collection of font families.

FontCollectionExtensions
: Extension methods for IFontCollection.

FontDescription
: Provides basic descriptive metadata for the font.

FontException
: Base class for exceptions thrown by this library.

FontFamilyNotFoundException
: Exception for detailing missing font families.

FontGlyphMetrics
: Represents a glyph metric from a particular font face.

FontMetrics
: Represents a font face with metrics, which is a set of glyphs with a specific style (regular, italic, bold etc).

FontPalette
: Selects the CPAL color palette used to resolve COLR glyph colors and optionally replaces individual palette entries.

  A palette index outside the range defined by the font selects the default palette (index 0), and overrides still apply, matching the CSS `font-palette` behavior. Overrides apply in order, so a later override of the same entry replaces an earlier one, and overrides whose entry index lies outside the font's palette entry range are ignored.

GlyphMissingException
: Exception for detailing missing font families.

GlyphOptions

Provides configuration options for rendering a glyph by id.

HorizontalMetrics
Represent the metrics of a font face specific to horizontal text.

InvalidFontFileException
Exception font loading can throw if it encounters invalid data during font loading.

InvalidFontTableException
Exception font loading can throw if it encounters invalid data during font loading.

KnownVariationAxes
Defines the registered design-variation axis tags for variable fonts. These tags are used with FontVariation to control font design axes. https://learn.microsoft.com/en-us/typography/opentype/spec/dvaraxisreg

LayoutModeExtensions
Extensions to LayoutMode.

LineLayout
Represents one laid-out line from a TextBlock.

LineLayoutEnumerator
Walks a TextBlock one laid-out line at a time.

MissingFontTableException
Exception font loading can throw if it finds a required table is missing during font loading.

SystemFontFallbackResolver
Resolves fallback font families from the fonts installed on the current machine using the operating system's character-to-font matching service. Results depend on the machine's installed fonts: the same input can resolve different families, or none, on different machines.

SystemFonts
Provides a collection of fonts.

TextBlock
Represents text prepared for repeated line layout, measurement, and rendering.

TextDecorationOptions
Represents overrides for the geometry of a single text decoration line, allowing a TextRun to replace the values that would otherwise be derived from the font metrics.

TextMeasurer
Encapsulates logic for laying out and then measuring text properties.

TextMetrics
: Encapsulates the full set of measurement results for laid-out text.

TextOptions
: Provides configuration options for rendering and shaping text.

TextRun
: Represents a run of text spanning a series of graphemes within a string.

TextShaper
: Encapsulates logic for shaping text into a positioned glyph stream.

TextShapingBuffer
: A reusable, caller-owned buffer holding text, the properties it is shaped under, and the glyphs shaping produces. Text and properties are set on the buffer, the buffer is shaped, and the glyphs are read back from it. Each shaping call replaces the glyphs.

VerticalMetrics
: Represent the metrics of a font face specific to vertical text.

Structs

CaretPosition
: Represents a caret line in laid-out text.

ClipBounds
: Represents a rectangular clipping region as axis-aligned bounds in the design space of the glyph source together with the transform that maps that space to the rendering surface.

FontFamily
: Defines a group of type faces having a similar basic design and certain variations in styles.

FontMatch
: Represents a font family and style matched for a character.

FontPaletteOverride
: Replaces the color of a single palette entry when a FontPalette is applied. An override whose Index lies outside the font's palette entry range is ignored.

FontRectangle
: Stores a set of four single precision floating points that represent the location and size of a rectangle.

FontVariation

Represents a single variation axis setting for a variable font, consisting of a four-character tag and a value.

Glyph
Represents a font-specific glyph at the point size used for layout and rendering.

GlyphColor
Provides access to the color details for the current glyph.

GlyphMetrics
Represents one laid-out glyph entry in final layout order.

GraphemeMetrics
Represents one coalesced grapheme in final layout order.

LineMetrics
Encapsulates measured metrics for a single laid-out text line.

ShapedGlyph
Represents a single shaped glyph: the result of substitution and positioning, before line breaking.

TextHit
Represents a hit-tested grapheme position in laid-out text.

TextPlaceholder
Represents an atomic inline placeholder with caller-supplied dimensions.

WordMetrics
Represents the positioned metrics for one Unicode word-boundary segment.

Interfaces

IFontCollection
A readable and writable collection of fonts.

IFontFallbackResolver
Resolves a font family for code points that no configured font can shape. The shaping pipeline consults the resolver only after Font and every FallbackFontFamilies entry have attempted the text, and at most once per distinct unresolved code point per shaping operation.

IMetricsHeader
Defines the contract for the metrics header of a font face.

IReadOnlyFontCollection

Represents a readonly collection of fonts.

IReadOnlySystemFontCollection
Represents a readonly collection of Operating System fonts.

Enums

CaretMovement
Specifies a caret movement operation within laid-out text.

CaretPlacement
Specifies an absolute caret placement within a laid-out text scope.

ColorFontSupport
Specifies which color font formats are enabled for layout and rendering.

DecorationPositioningMode
Defines how text decorations (underline, overline, strikethrough) are positioned relative to font metrics.

FontStyle
The font styles

FontWeight
Identifies the visual weight of a font on the OpenType and CSS weight scale.

GlyphLayoutMode
Provides enumeration for the various layout mode of an individual glyph within a body of text.

GlyphType
Represents the various versions of a glyph records.

HintingMode
Defines modes to determine how to apply hinting. The use of mathematical instructions to adjust the display of an outline font so that it lines up with a rasterized grid.

HorizontalAlignment
Horizontal alignment modes.

KerningMode
Kerning is the contextual adjustment of inter-glyph spacing. This property controls metric kerning, kerning that utilizes adjustment data contained in the font.

LayoutMode
Defines modes to determine the layout direction of text.

TextAlignment
: Text alignment modes.

TextAttributes
: Provides enumeration of various text attributes.

TextBaseline
: Specifies which reference line of laid-out text is placed at Origin along the block flow axis, and equivalently which line rides the path when text follows one. Values match the CSS `dominant-baseline` and HTML canvas `textBaseline` vocabulary.

TextBidiMode
: Specifies how bidirectional text is resolved.

TextDecorationSkipInk
: Identifies whether underline and overline decorations are interrupted where they would otherwise cross glyph ink, mirroring the CSS `text-decoration-skip-ink` property. Strikethrough decorations always cross the ink regardless of this setting.

TextDecorations
: Provides enumeration of various text decorations.

TextDirection
: Specifies the writing direction for text.

TextEllipsis
: Specifies ellipsis behavior when laid-out text is limited to a maximum number of lines.

TextHyphenation
: Specifies hyphenation marker behavior when text breaks at hyphenation opportunities.

TextInteractionMode
: Specifies how text interaction positions are modeled for laid-out text.

TextJustification
: Text justification modes.

TextPlaceholderAlignment
: Specifies how an inline placeholder is aligned against surrounding text.

VerticalAlignment
: Vertical alignment modes.

WordBreaking

Defines modes to determine when line breaks should appear when words overflow their content box.

Namespace SixLabors.PolygonClipper

Classes

Contour
: Represents a single polygon ring (outer contour or hole).

Polygon
: Represents a complex polygon.

PolygonClipper
: Performs boolean operations on polygons.

PolygonStroker
: Generates polygonal stroke geometry for contours with configurable joins and caps.

StrokeOptions
: Provides configuration options for geometric stroke generation.

Structs

Box2
: Represents a bounding box.

Vertex
: Represents a two-dimensional vertex with X and Y coordinates.

Enums

BooleanOperation
: Specifies the type of boolean operation to perform on polygons.

LineCap
: Specifies the shape to be used at the ends of open lines or paths when stroking.

LineJoin
: Specifies how the connection between two consecutive line segments (a join) is rendered when stroking paths or polygons.

ResultTransition
: Represents the result transition for a sweep event.